

Balanced Simplex Joins Interpolate the Vertex Endpoints

Candidate partial result for arXiv:2509.13898

Model: GPT5.6

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Status: partial result, likely valid. For every $1 \leq r \leq n$ we construct an n -polytope with exactly $n + r$ vertices and isoperimetric quotient $\asymp n/\sqrt{r}$. The matching universal lower bound remains open.

1 Source problem

For a convex body $K \subset \mathbb{R}^n$, write

$$\text{iq}(K) = \frac{\text{vol}_{n-1}(\partial K)}{\text{vol}_n(K)^{(n-1)/n}}.$$

Ball, Böröczky, and Naor ask how the optimal universal lower bound on $\text{iq}(K)$ interpolates when an n -dimensional polytope has β vertices with $n + 1 < \beta < 2n$ [1, Remark 2]. The two endpoints are represented by the regular simplex, with quotient of order n , and the cross-polytope, with quotient of order $\sqrt{n}$.

Remark 2. One could also wonder about improving the classical estimate $\text{iq}(K) \gtrsim \sqrt{n}$ in [2] when K is restricted to be a convex polytope that has β vertices for some integer $\beta \geq n + 1$. If $\beta \geq 2n$, then this is impossible since $B_{\ell_1^n}$ has $2n$ vertices and $\text{iq}(B_{\ell_1^n}) \asymp \sqrt{n}$, so by considering the convex hull of $B_{\ell_1^n}$ with additional $\beta - 2n$ points from $\mathbb{R}^n \setminus B_{\ell_1^n}$ that are in general position and arbitrarily close to $\partial B_{\ell_1^n}$, one sees that there is a convex polytope in $\mathbb{R}^n$ that has β vertices and whose isoperimetric quotient is of order $\sqrt{n}$. If $\beta = n + 1$, then a stronger isoperimetric lower bound holds since a convex polytope $K \subseteq \mathbb{R}^n$ that has $n + 1$ vertices is a simplex, whence $\text{iq}(K) \geq \text{iq}(\Delta_n) \asymp n$ by [26] (this also follows from [36]), where $\Delta_n \subseteq \mathbb{R}^n$ is the regular simplex. It remains open to determine how to interpolate between the aforementioned asymptotic lower bounds on $\text{iq}(K)$ when K is a convex polytope that has β vertices and $n + 1 < \beta < 2n$.

More generally, one could ask how the estimate $\text{iq}(K) \gtrsim \sqrt{n}$ in [2] improves when K is a convex polytope that has a fixed number of faces of a given intermediate dimension, or even while fixing its numbers of faces whose dimensions belong to given subset of $\{0, \dots, n - 1\}$; to the best of our knowledge, this question has not been broached in the literature.

Figure 1: Remark 2 on page 2 of the source paper.

To state the contribution compactly, define

$$\mathcal{I}_n(\beta) = \inf\{\text{iq}(P) : P \subset \mathbb{R}^n \text{ is a convex polytope with exactly } \beta \text{ vertices}\}.$$

2 Idea of the proof

The endpoint bodies are themselves joins of regular simplices: one n -simplex at $r = 1$, and the join of n one-dimensional simplices at $r = n$, which is the cross-polytope. The natural interpolation is

therefore to join r regular simplices whose dimensions are as equal as possible.

The source paper already computes the volume and surface area of an arbitrary join (called an ℓ_1 -sum there) [1, Lemma 12, equations (24)–(26)]. Scaling a b -simplex to inradius $1/\sqrt{b}$ makes the surface factor from that lemma add exactly as $\sqrt{b_1 + \dots + b_r} = \sqrt{n}$. What remains is a single balanced product. Stirling's formula turns that product into $\sqrt{n/r}$, yielding $n/\sqrt{r}$ after the surface prefactor is restored.

3 The partial result

Theorem 1 (Balanced simplex-join interpolation). *There are universal constants $c, C > 0$ such that for every $n \geq 1$ and every integer r with $1 \leq r \leq n$, there exists an n -dimensional convex polytope $K_{n,r} \subset \mathbb{R}^n$ with exactly $n + r$ vertices and*

$$c \frac{n}{\sqrt{r}} \leq \text{iq}(K_{n,r}) \leq C \frac{n}{\sqrt{r}}.$$

In particular,

$$\mathcal{I}_n(n + r) \leq C \frac{n}{\sqrt{r}}.$$

Proof. Choose positive integers $b_1, \dots, b_r$ with

$$b_1 + \dots + b_r = n, \quad b_i \in \{\lfloor n/r \rfloor, \lceil n/r \rceil\}.$$

For each i , let $\Delta_i \subset \mathbb{R}^{b_i}$ be a regular simplex, centered at the origin and scaled so that its inradius is $h_i = 1/\sqrt{b_i}$. In mutually orthogonal coordinate blocks, form their join

$$K_{n,r} = \text{conv}(\Delta'_1 \cup \dots \cup \Delta'_r) \subset \mathbb{R}^{b_1} \times \dots \times \mathbb{R}^{b_r} = \mathbb{R}^n.$$

This is precisely the ℓ_1 -sum in equation (24) of [1]. The vertex count in Lemma 12 of that paper gives

$$\# \text{vert}(K_{n,r}) = \sum_{i=1}^r (b_i + 1) = n + r.$$

We next record the simplex normalization. A regular b -simplex of side length a has volume

$$\frac{a^b \sqrt{b+1}}{b! \sqrt{2^b}} \quad \text{and inradius} \quad \frac{a}{\sqrt{2b(b+1)}}.$$

Consequently, a regular b -simplex with inradius h has volume

$$\frac{h^b b^{b/2} (b+1)^{(b+1)/2}}{b!}.$$

At $h = 1/\sqrt{b}$ this becomes

$$b! \text{ vol}_b(\Delta_i) = (b+1)^{(b+1)/2}. \tag{1}$$

Let

$$P = \prod_{i=1}^r (b_i + 1)^{(b_i+1)/2}.$$

The volume and surface formulas (25) and (26) of [1], together with (1) and $\sum_i h_i^{-2} = \sum_i b_i = n$, give

$$\text{vol}_n(K_{n,r}) = \frac{P}{n!}, \quad \text{vol}_{n-1}(\partial K_{n,r}) = \frac{\sqrt{n} P}{(n-1)!} = n\sqrt{n} \text{vol}_n(K_{n,r}).$$

It follows exactly that

$$\text{iq}(K_{n,r}) = n\sqrt{n} \left(\frac{P}{n!} \right)^{1/n}. \quad (2)$$

Put $m = n/r \geq 1$. Balance of the blocks implies

$$m \leq b_i + 1 \leq m + 2 \leq 3m, \quad \sum_{i=1}^r (b_i + 1) = n + r = n \left(1 + \frac{1}{m} \right).$$

Therefore

$$\sqrt{m} \leq P^{1/n} \leq (3m)^{(1+1/m)/2} \leq 3\sqrt{m},$$

where the last constant is universal. Stirling's formula gives $(n!)^{1/n} \asymp n$. Substitution into (2) yields

$$\text{iq}(K_{n,r}) \asymp \sqrt{n} \sqrt{m} = \frac{n}{\sqrt{r}},$$

as claimed. □

Corollary 2 (Current bracket for the open interpolation). *For $1 \leq r \leq n$,*

$$c\sqrt{n} \leq \mathcal{I}_n(n+r) \leq C \frac{n}{\sqrt{r}}.$$

The lower estimate is the classical Euclidean isoperimetric inequality.

4 Verification notes

Three checks are especially diagnostic.

1. If $r = 1$, then $b_1 = n$ and (2) is the regular-simplex formula in equation (11) of [1], asymptotic to en .
2. If $r = n$, then every $b_i = 1$ and the join is the cross-polytope. Equation (2) becomes $2n\sqrt{n}/(n!)^{1/n} \sim 2e\sqrt{n}$, again matching equation (11).
3. The first part of Lemma 12, namely equations (24)–(26), does not require origin symmetry. Origin symmetry is imposed only for the later affine-minimality formula (27), which is not used here.

The script `code/verify_formula.py` evaluates the logarithmic form of (2) for all $1 \leq r \leq n \leq 500$. This finite computation is only a consistency audit; the proof above is uniform and noncomputational.

5 Limitations and novelty check

The theorem supplies an interpolating *upper construction* for the extremal infimum. It does not establish the conjecturally matching statement that every n -polytope with $n + r$ vertices satisfies $\text{iq}(K) \gtrsim n/\sqrt{r}$. Thus Remark 2 remains open.

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for the source id and the vertex-interpolation terminology. Exact web/arXiv searches for “ $n + r$ vertices,” “isoperimetric quotient,” “join,” and the scale $n/\sqrt{r}$ found the source paper but no occurrence of this construction. This was a bounded search rather than a complete MathSciNet or zbMATH audit, so the novelty assessment is provisional. The source uses Lemma 12 for symmetric cube blocks in the regime of at least $2n$ vertices; it does not state the balanced-simplex application above.

References

- [1] K. Ball, K. J. Böröczky, and A. Naor, *Approximate isoperimetry for convex polytopes*, arXiv:2509.13898 (2025), especially Remark 2 and Lemma 12.